Group 2

Trends: Physical properties

Properties	Trends
atomic radii/ ionic radii	__________ down the group ❶ shielding effect (no. of e^- shell) ❷ nuclear charge ❸ electrostatic attraction between nucleus and <u>outermost e^-</u> ⇒ e^- cloud size
1st I.E.	__________ down the group ❶ shielding effect (no. of e^- shell) ❷ nuclear charge ❸ electrostatic attraction between nucleus and <u>outermost e^-</u> ⇒ energy req. to remove outermost e^-
electronegativity	__________ down the group ❶ shielding effect (no. of e^- shell) ❷ nuclear charge ❸ electrostatic attraction between nucleus and <u>bonding e^-</u>

Trends: Chemical properties

Properties	Trends
reactivity of metal (M)	________ down the group

<u>$E^\ominus$ value from Data Booklet:</u>

$Mg^{2+} + 2e^- \rightleftharpoons Mg$ $E^\ominus = -2.38$ V
$Ca^{2+} + 2e^- \rightleftharpoons Ca$ $E^\ominus = -2.87$ V
$Ba^{2+} + 2e^- \rightleftharpoons Ba$ $E^\ominus = -2.90$ V

How does trend in $E^\ominus$ value explain reactivity?

Down the group,

- $E^\ominus$ value becomes more __________
- tendency for ____________________ reaction increases
- tendency of metal losing e^- ___________
- reducing power of metal ___________
- reactivity of metal ______________

Carbonates: Thermal stability

Properties	Trends
decomposition temperature	$MCO_3 \rightarrow MO + CO_2$ increases down the group

illustration:

Explain why does thermal stability of Group 2 carbonates increase down the group.

Down the group,

- cationic radii __________
- charge density of cation __________
- polarising power of cation __________
- extent of distortion on electron cloud of CO_3^{2-} __________
- extent of weakening of covalent bond in CO_3^{2-} __________
- energy req to break the covalent bonds ________________

Group 17

Trends: Physical properties

Properties	Trends
atomic radii/ ionic radii	__________ down the group
1st I.E.	__________ down the group
electronegativity	__________ down the group
bond dissociation energy of X_2	__________ down the group • diffuse-ness of valence orbital • effectiveness of orbital overlap
volatility of X_2	__________ down the group • size and polarisability of electron cloud • strength of id-id • energy req to overcome

Trends: Chemical properties

Properties	Trends
reactivity of halogen (X_2)	________ down the group

<u>$E^\ominus$ value from Data Booklet:</u>

$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$ $E^\ominus = +1.36$ V
$Br_2 + 2e^- \rightleftharpoons 2Br^-$ $E^\ominus = +1.07$ V
$I_2 + 2e^- \rightleftharpoons 2I^-$ $E^\ominus = +0.54$ V

How does trend in $E^\ominus$ value explain reactivity?

Down the group,

- $E^\ominus$ value becomes less ___________
- tendency for ____________________ reaction decreases
- tendency of halogen gaining e^- _________
- oxidising power of halogen __________
- reactivity of halogen ____________

HX: Thermal stability

Properties	Trends
decomposition temperature	$HX \rightarrow \frac{1}{2} H_2 + \frac{1}{2} X_2$ decreases down the group

Note: HF and HCl do not decompose on heating

Explain why does thermal stability of hydrogen halides decrease down the group.

Down the group,

1. the valence p orbital of the halogen becomes ____________________. Hence there is ____________________ overlap between the 1s orbital of hydrogen and the valence p orbital of halogen.
2. the electronegativity difference between hydrogen and halogen ___________, resulting in a ______________ bond.

Hence, H-X bond strength ___________ down the group

<u>Displacement reactions</u>

	Cl^-	Br^-	I^-
Cl_2	–		
Br_2		–	
I_2			–

Notes:

- more reactive halogen displaces less reactive halide

<u>Reaction with thiosulfate</u>

Cl_2 & Br_2	Oxidises $S_2O_3^{2-}$ to ______
I_2	Oxidises $S_2O_3^{2-}$ to ______

Write half eq^n then combine:

[O]:
[R]:
Overall :

<u>Reaction with Fe</u>

Cl_2 & Br_2	Oxidises Fe to ________
I_2	Oxidises Fe to ________

Write half eq^n then combine:

[O]:
[R]:
Overall :

Test for X^-

Test	Observation & identity
Step 1: Step 2:	For Cl^-: For Br^-: For I^-:

Group 2

Trends: Physical properties

Properties	Trends
atomic radii/ ionic radii	increases down the group ❶ shielding effect (no. of e^- shell) ❷ nuclear charge ❸ electrostatic attraction between nucleus and outermost e^- ⇒ e^- cloud size
1^{st} I.E.	decreases down the group ❶ shielding effect (no. of e^- shell) ❷ nuclear charge ❸ electrostatic attraction between nucleus and outermost e^- ⇒ energy req. to remove outermost e^-
electronegativity	decreases down the group ❶ shielding effect (no. of e^- shell) ❷ nuclear charge ❸ electrostatic attraction between nucleus and bonding e^-

Trends: Chemical properties

Properties	Trends
reactivity of metal (M)	increases down the group

$E^\ominus$ value from Data Booklet:

$Mg^{2+} + 2e^- \rightleftharpoons Mg$ $\quad E^\ominus = -2.38$ V
$Ca^{2+} + 2e^- \rightleftharpoons Ca$ $\quad E^\ominus = -2.87$ V
$Ba^{2+} + 2e^- \rightleftharpoons Ba$ $\quad E^\ominus = -2.90$ V

How does trend in $E^\ominus$ value explain reactivity?

Down the group,

- $E^\ominus$ value becomes more negative
- tendency for backward oxidation reaction increases
- tendency of metal losing e^- increases
- reducing power of metal increases
- reactivity of metal increases

Carbonates: Thermal stability

Properties	Trends
decomposition temperature	$MCO_3 \rightarrow MO + CO_2$ increases down the group

illustration:

Explain why does thermal stability of Group 2 carbonates increase down the group.

Down the group,

- cationic radii increases
- charge density of cation decreases
- polarising power of cation decreases
- extent of distortion on electron cloud of CO_3^{2-} decreases
- extent of weakening of covalent bond in CO_3^{2-} decreases
- energy req to break the covalent bonds increases

Group 17

Trends: Physical properties

Properties	Trends
atomic radii/ ionic radii	increases down the group
1^{st} I.E.	decreases down the group
electronegativity	decreases down the group
bond dissociation energy of X_2	decreases down the group • diffuse-ness of valence orbital • effectiveness of orbital overlap
volatility of X_2	decreases down the group • size and polarisability of electron cloud • strength of id-id • energy req to overcome intermolecular force of attraction

Trends: Chemical properties

Properties	Trends
reactivity of halogen (X_2)	decreases down the group

$E^\ominus$ value from Data Booklet:

$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$ $\quad E^\ominus = +1.36$ V
$Br_2 + 2e^- \rightleftharpoons 2Br^-$ $\quad E^\ominus = +1.07$ V
$I_2 + 2e^- \rightleftharpoons 2I^-$ $\quad E^\ominus = +0.54$ V

How does trend in $E^\ominus$ value explain reactivity?

Down the group,

- $E^\ominus$ value becomes less positive
- tendency for forward reduction reaction decreases
- tendency of halogen gaining e^- decreases
- oxidising power of halogen decreases
- reactivity of halogen decreases

HX: Thermal stability

Properties	Trends
decomposition temperature	$HX \rightarrow \frac{1}{2} H_2 + \frac{1}{2} X_2$ decreases down the group

Note: HF and HCl do not decompose on heating

Explain why does thermal stability of hydrogen halides decrease down the group.

Down the group,

1. the valence p orbital of the halogen becomes more diffuse. Hence there is less effective overlap between the 1s orbital of hydrogen and the valence p orbital of halogen.
2. the electronegativity difference between hydrogen and halogen decreases, resulting in a less polar bond.

Hence, H-X bond strength decreases down the group

Displacement reactions

	Cl^-	Br^-	I^-
Cl_2	–	$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$	$Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$
Br_2	no reaction	–	$Br_2 + 2I^- \rightarrow 2Br^- + I_2$
I_2	no reaction	no reaction	–

Notes:

- more reactive halogen displaces less reactive halide

Reaction with thiosulfate

Cl_2 & Br_2	Oxidises $S_2O_3^{2-}$ to SO_4^{2-}
I_2	Oxidises $S_2O_3^{2-}$ to $S_4O_6^{2-}$

Write half eq^n then combine:

[O]: $2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2e^-$
[R]: $I_2 + 2e^- \rightarrow 2I^-$
Overall : $2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$

Reaction with Fe

Cl_2 & Br_2	Oxidises Fe to Fe^{3+}
I_2	Oxidises Fe to Fe^{2+}

Write half eq^n then combine:

[O]: $Fe \rightarrow Fe^{2+} + 2e^-$
[R]: $I_2 + 2e^- \rightarrow 2I^-$
Overall : $Fe + I_2 \rightarrow Fe^{2+} + 2I^-$

Test for X^-

Test	Observation & identity
Step 1: add $AgNO_3$(aq) Step 2: add excess NH_3(aq) OR add conc. NH_3	For Cl^-: white ppt of AgCl formed, soluble in excess NH_3(aq) For Br^-: cream ppt of AgBr formed, soluble in conc NH_3 For I^-: yellow ppt of AgI formed, insoluble in conc NH_3